

Prime Parts Assembly Order and Nucleon Readouts

A finite WRRR case study of integer address 105

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Abstract and assessment path

Verified inputs The published assembler and recorded nucleon/electron rest energies are inherited. Address 105, the first 15 primes, two traversal orders and the readout assignment are fixed configuration choices.

WRRR transformation Assemble the address from prime-labelled components, retain their identities during nucleon readout, and compare both assignments under a common source budget including unused components and energy differences.

Outputs Small-first gives 11 components, read as 1 neutron n and 10 D; large-first gives 4, read as 1 n and 3 D. D is a proton/electron/electron-antineutrino bundle. Both retain B=11 and energy 10335.21964134 MeV.

Falsifiers A different result under the fixed rules, or failure of the address identity, charge, baryon, lepton or total-energy balance, falsifies this executed construction.

1 Finite assembly rule

This case presents the assembly-to-nucleon connection of the WRRR prime-parts series [1]. A prime is a component label; an integer address is their additive sum plus any residue. The pool contains the first 15 primes.

$$2, 3, 5, 7, 11, 13, 17, 19, 23, 29, 31, 37, 41, 43, 47$$

Start with remainder 105. Visit each prime once in the chosen order; if it fits, select one component and subtract its label. After a full traversal, repeat the order while the remainder is at least 2. Stop at residue 0 or 1.

2 Two assemblies of the same address

$$105 = 2 + 3 + 5 + 7 + 11 + 13 + 17 + 19 + 23 + 2 + 3$$
$$105 = 47 + 43 + 13 + 2$$

Quantity	Small-first	Large-first
Active components	11	4
Neutron component label	5	47
Nucleon readout	1 n + 10 D	1 n + 3 D
Unused source baryons	0	7
Address residue	0	0

Small-first consumes 100 in its first traversal and then selects 2 and 3. Large-first selects 47 and 43, then the fitting 13 and 2. Different active configurations therefore correspond to the same additive address.

3 Nucleon readout and a shared source budget

Labels 5 and 47 read as neutron n ; other selected labels read as a complete D bundle of proton, electron and electron antineutrino. Each component has charge $Q=0$, baryon number $B=1$ and lepton number $L=0$. Electron and antineutrino lepton numbers cancel. Prime labels and occurrence indices are retained.

The common source contains 11 neutral $B=1$ components. The seven unused in large-first retain neutron-like energy shells as inactive reserves. This is a preparation choice, not an observed free neutron rendered stable.

Recorded rest energies are 939.56542194 MeV (neutron), 938.27208943 MeV (proton) and 0.51099895069 MeV (electron). Proton-plus-electron rest energy is 938.78308838069 MeV per D. The remaining 0.78233355931 MeV is an unresolved antineutrino/kinetic/recoil budget. Individual spectra and photon production are not computed here.

Energy term in MeV	Small-first	Large-first
Active particle rest energy	10327.39630574690	3755.91468708207
D nonrest energy budget	7.82333559310	2.34700067793
Unused source energy	0	6576.95795358
Total and common source	10335.21964134000	10335.21964134000

Address 105 is an additive label. Both address residues are zero; the large-first source reserve still holds seven baryons. Address and physical-resource balances are separate.

4 Interpretation and external review

The calculation executes a WRRRA connection from assembly order to active nucleon configuration. Known rest energies and complete source accounting support the reality-consistency of this chosen construction. These are two assignments with common total budgets. Dynamics that produces both from the same microscopic state remains a question for review.

Review should examine how prime-dependent readouts and inactive reserves can be physically prepared, and which changes persist when pool, order and assignment vary. This dataset executes static assembly, readout and source accounting.

5 Separate dataset and reproduction

Dataset [2] contains input JSON, 45 traversal visits, 15 selected components, 30 per-prime records, two ledgers, results and a standard-library Python reproducer. Its 28 checks and 13 upstream comparisons pass; these are implementation/conservation checks. Run `python reproduce.py --check` for reference-byte comparison.

Data and references

[1] Wonsik Choi. WRRRA Prime Parts Reviewed Collection 0.1-0.12. DOI 10.5281/zenodo.23134716.

[2] Wonsik Choi. WRRRA Address 105 Prime Parts Case Study Dataset, 1.0.0. DOI 10.5281/zenodo.23137540. GitHub case paper and dataset